

THE ECONOMIC EFFECT OF COVID LOCKDOWN POLICIES

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ABSTRACT. This paper presents a mathematical model of the COVID-19 pandemic. We extend the standard SIR model by including two new states, exposed (E) and death (D), denoting this model as the SEIRD model. The model also includes additional transition edges for vaccination and virus mutation. We augment the model dynamics with a COVID lockdown function, parameterized by s and m . The parameter s controls the speed of a lockdown, and m controls the strength. We then present a total cost objective function that balances total deaths, economic cost, and a hospital over-capacity penalty. We follow by modifying the economic cost to account for individual-specific productivity levels. We present the optimal lockdown parameter set and conduct a sensitivity analysis for the most important parameters.

Keywords. Differential equations, COVID lockdown policy, mathematical modeling, epidemiology, optimization

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1. INTRODUCTION

Although the COVID-19 pandemic is behind us in the year 2025, it remains worthwhile to analyze to prepare for future pandemic events. For this paper, we focus on lockdown policies and examine the interplay between different policies and three cost measures: death, economic cost, and hospital overcapacity.

We make the simplifying assumption that a lockdown policy is explained by two parameters: speed and strength. We allow the lockdown policy to vary in this way to capture the heterogeneity in speed and severity of lockdown policies. During the pandemic, some states operated via a gradual lockdown, starting with bans on large gatherings, moving to recommendations to work from home, then a stay-at-home mandate (e.g., Massachusetts [24]). Others were more strict and swift, mandating a complete lockdown from day one (e.g., California [19]).

The deterministic SIR model provides a strong baseline for modeling the COVID pandemic AlQadi and Bani-Yaghoub [4]. The model captures key characteristics, such as herd immunity, peak infection, and the dynamics of growth and slowdown in an epidemic. However, the SIR model is limited by three states: Susceptible (S), Infected (I), and Recovered (R). The parameters in the model are also restricted to be time-invariant, which may be too inflexible. The transmission rate a , for example, changes based on lockdown policy or the environment. As such, we extend the SIR framework to include two new states exposed (E) and death (D), and modify the transmission rate $a(t)$ to be time varying.

2. DESCRIPTION OF THE MODEL

The proposed SEIRD model (Beckley et al.) is specified by the following system of ordinary differential equations (ODE's):

$$\frac{dS}{dt} = -\frac{a(t)}{N}IS - cS + \gamma R \quad (2.1)$$

$$\frac{dE}{dt} = \frac{a(t)}{N}IS - \nu E \quad (2.2)$$

$$\frac{dI}{dt} = \nu E - bI - \eta I \quad (2.3)$$

$$\frac{dR}{dt} = bI + cS - \gamma R \quad (2.4)$$

$$\frac{dD}{dt} = \eta I. \quad (2.5)$$

The units are individuals and time t is in days. The five possible states an individual can move to are denoted as S , E , I , R , and D (Susceptible, Exposed, Infected, Recovered and Death, respectively). The inclusion of E was to model the latent period of COVID. When an individual is infected with COVID, they don't instantaneously become infectious. It takes time before they can transmit the virus to others. D is the terminal state of death, included to relax the SIR model assumption that all infected individuals recover. For the entirety of this project, our initial condition assumes the patient zero scenario, starting with just one exposed individual and all others as susceptible.

In addition to the two new states, we include two new parameters c and γ , vaccination rate and mutation rate, respectively (Smith [23]). If a susceptible individual is vaccinated, they move to the recovered state. However, some recovered individuals go back to the susceptible state because, as the virus mutates, the initial vaccine becomes ineffective. The values for c and γ are fixed and were determined by Mathieu et al. [17] and Manica et al. [14]. The values given in the papers were converted to match our units. Figure 1 illustrates the graph for the final SEIRD model, where each state is a node and rates are edges.

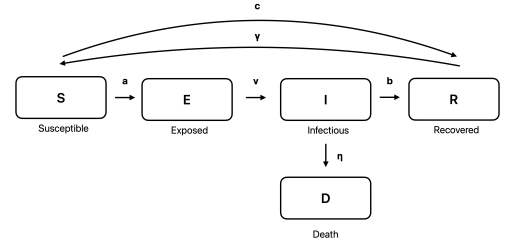


FIGURE 1. SEIRD Model Graph

All other parameters are carried over from the SIR model. N denotes population. In our system, the population is closed (no births or deaths) fixed at $2e6$ to model a large city or medium-sized metropolitan statistical area (MSA). b is the recovery rate from Oelsner et al. [18]. ν is incubation rate. η is the death rate given by the data from Mathieu et al. [17]. To convert from the case fatality rate (CFR) given in the data to the death rate used in our analysis, we use the following formula and solve for η :

$$CFR = \frac{\eta}{\eta + b}. \quad (2.6)$$

All parameter units are in units of per person per day.

Finally, we modify the transmission rate to be time-varying, denoted as $a(t)$. The parameter can be decomposed as a product of the probability of transmission (pt) and the number of contacts, $nc(t)$:

$$a(t) = pt \cdot nc(t). \quad (2.7)$$

$nc(t)$ is time-dependent due to lockdown and is modulated by two parameters s and m_{min} . It is described by the following function:

$$nc(t) = m_{min} + (m_{max} - m_{min}) \cdot \exp[s \cdot \max(t - t_{start}, 0)]. \quad (2.8)$$

We define lockdown policy as a specific pair of (s^*, m_{min}^*) parameter values, where $\{s \in \mathbb{R} \mid s \geq 0\}$ and $\{m_{min} \in \mathbb{R} \mid m_{min} \in [1, 5]\}$.

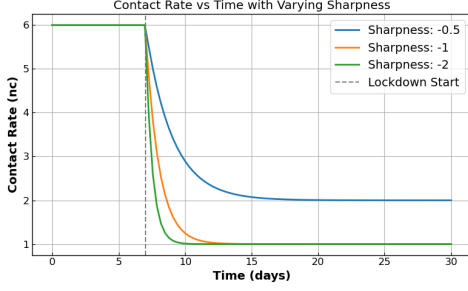
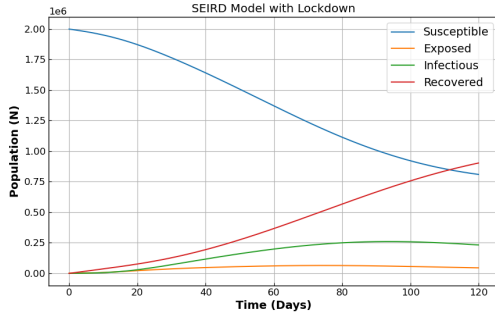


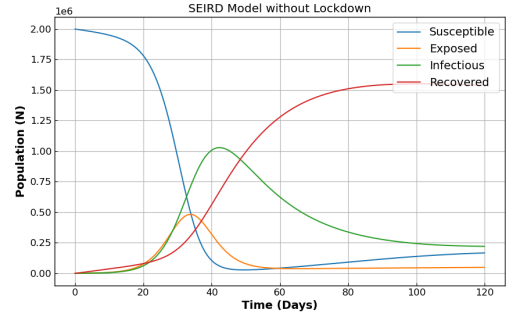
FIGURE 2. Lockdown Function

For equation 2.7, we bound the number of contacts between 6 and 1. A lower bound of 1 makes sense because people at a minimum need to interact with one other person to survive (e.g., buy food, get water). A value of 6 is also reasonable because the average number of close friends is around 5 using Dunbar's number (Scotch [22]).

3. ANALYSIS



(A)



(B)

FIGURE 3. Solution Curves: Lockdown vs No Lockdown

In Figure 3, we present the solution curves for the system of ODEs (equations 2.1 - 2.5). Figure 3(A) shows the solution curves with a lockdown present, while Figure 3(B) presents solution curves in the absence of a lockdown. The lockdown has the effect of flattening the curves for the infected and exposed groups. We see the susceptible group decline and the recovered group increase because of vaccines. The peaks for infected and exposed are much more pronounced without lockdown. We see a lag between exposed and infected, a property of the SEIRD model. Note, cumulative death is omitted from the above figures for visual clarity, but can be found in Figure 4(B).

The optimization problem is to minimize total cost by modifying decision variables s and m_{min} to find the optimal policy (s^*, m_{min}^*) . Mathematically, we express it as:

$$\begin{aligned} \min_{s, m_{min}} \quad & TC(s, m_{min}) \\ \text{s.t.} \quad & \end{aligned} \tag{3.1}$$

$$\begin{aligned} \text{SEIRD Model with Lockdown } nc(t, s, m_{min}) \\ s \leq 0, \end{aligned} \tag{3.2}$$

$$m_{min} \in [1, 5],$$

where:

$$TC(s, m_{min}) = V_{life} \cdot D(t) + GDP_{day} \int_0^T nc(t, s, m_{min}) dt + C_{hosp} \int_0^T \max(I(t) - H_{cap}, 0) dt \quad (3.3)$$

The first term is the cost of deaths, which is the product $V_{life} \cdot D(t)$. V_{life} is the value of life in millions of dollars, and $D(t)$ is the cumulative number of deaths at the end of the period $[0, T]$. V_{life} was set to be \$15M because the Department of Transportation [8] finds it to be \$13.7M in 2024 or around \$14M in 2025 dollars. A sensitivity analysis on this parameter is conducted in Section 3.2 to ensure results are robust. The second term is the economic cost. To obtain this value, we first normalize $nc(t)$ to be between 0 and 1. This value represents the fraction of GDP lost during lockdown. We then integrate it over $[0, T]$ to get the total lockdown days, weighed by the economic impact GDP_{day} . The implicit assumption is that lockdowns diminish productivity agnostic to the number of individuals in each SEIRD state. This assumption will be relaxed in Section 3.1. Finally, we have the hospital overflow penalty. The integral gives the total number of overflows, and we multiply it by the cost per day per patient exceeding capacity. All cost terms are equally weighted, and units are in millions of dollars.

H_{cap} is a fixed parameter that is a proportion of N . We use data (“Fast Facts on U.S. Hospitals, 2025 — AHA”) to estimate the average hospital beds in the U.S., which we find to be around 0.3% of N . For C_{hosp} , we use \$15k, derived from Kapinos et al. [11].

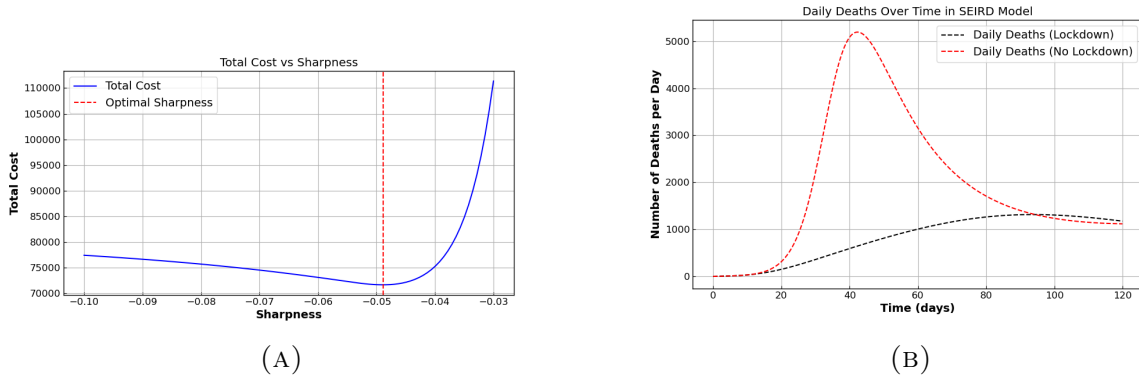


FIGURE 4. Optimal Sharpness and Daily Deaths

The optimum for the s is an interior point, as shown in Figure 4(A), while the optimum for m_{min} is on the boundary at 1. Many different parameterizations were attempted, and in all cases, m_{min} did not change. This may be due to the correlation of parameters or because it is always best to select the strongest lockdown. The latter makes sense because if one opts for a lockdown, it’s best to make it strong to eradicate the virus.

3.1. Cumulative Productivity Dependent on Lockdown Strategy. Now we relax the assumption lockdowns diminish productivity agnostic to the number of individuals in each state. We adjust the model to consider the expected productivity values for each population group.

Calculating productivity is difficult to measure using one unit across different industries, even individuals within an industry have different output measurements. To navigate this, we assume a unit of productivity to be equal to the output generated by a healthy individual in a unit of time. In a fully healthy society, the number of units of productivity is equal to the number of people.

Next, we assign each population group in the model a productivity coefficient, resulting in the following total productivity calculation:

$$TP(t) = \sum_{i \in \{S, E, I, R, D\}} u_i \cdot P_i(t) \quad (3.4)$$

, where $TP(t)$ is a total productivity at a time t . u_i provides the productivity coefficient associated with the i state and $P_i(t)$ is the population of the i state at time t . Note that $i \in \{S, E, I, R, D\}$.

To evaluate different lockdown strategies, we evaluated three different scenarios: a sharp lockdown curve ($s = -0.1$), a no lockdown strategy, and a slow lockdown strategy ($s = -0.04$). All other variables in the SEIRD model were kept consistent across the three evaluations.

Both susceptible and recovered individuals are considered healthy and fully productive, so their productivity coefficient is 1. An infected person is considered a negative contribution to productivity, as they need care from other members in the population to recover. We set this coefficient to be -0.25, as we expect a healthy individual could care for up to 4 infected individuals. This number is in line with Oregon House Bill 2697 [20], which provides registered nurse/patient limits for hospitals.

We were unable to identify an individual's productivity decrease due to illness in our research, so we assume a value of 0.8 based on the reasoning that a person progressing through an illness is less effective. Lastly, a deceased individual contributes no productivity.

For the no lockdown strategy, we set the array of productivity coefficients as follows.

$$u_i^{(\text{No Lockdown})} = \{1, .8, -0.25, 1, 0\} \quad (3.5)$$

In a lockdown scenario, these values change. Gibbs, Mengel, and Siemroth (2023) [9] observed that a transition to fully remote work results in a productivity between 8 to 19 percent. A 10 percent decrease appears reasonable in this context. Considering that change for susceptible and recovered individuals, the array of productivity coefficients is modified to:

$$u_i^{(\text{Lockdown})} = \{.9, .8, -0.25, .9, 0\} \quad (3.6)$$

This lockdown productivity coefficients array will be the same for both a sharp and a slow lockdown. The population counts will differ between the two total productivity calculations, leading to a difference in the total productivity observed.

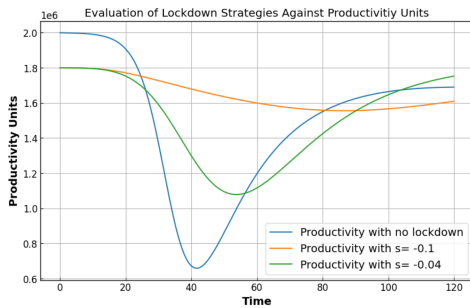


FIGURE 5. Productivity as a Result of Different Lockdown Strategies

no lockdown strategy. The results are shown in Figure 5.

To understand final productivity levels, we assume if there are no infections for 14 days, the lockdown is lifted, and all healthy populations return to full effectiveness. This period is based on the common isolation period defined by most governments during the pandemic, such as the U.S. Department of State's recommendation in March 2020 [25]. Although this was included in the model, none of the three evaluated strategies achieved pandemic elimination.

We then model the total productivity using these productivity coefficients and the population dynamics from the SEIRD model using a slow, sharp, and

This chart follows intuition. As the pandemic progresses, the lockdown strategies will result in higher productivity. This results from having fewer deaths during the pandemic and, therefore, more individuals able to contribute. A surprising observation is that during the pandemic, the no-lockdown strategy ended up having the lowest productivity. Intuition states that with more people working normally, productivity would exceed strategies incorporating a lockdown. However, this analysis suggests that the cost to treat infected individuals outweighs the benefit of people working in person. An interesting observation is that none of the models achieved pandemic elimination in the 120 day time period. This is because of the virus mutation parameter γ .

3.2. Sensitivity Analysis. We now consider the sensitivity of our result to changes in three parameters: death rate (η), probability of transmission (pt), and GDP per Day (GDP_{day}). We first examine the sensitivity of the optimal lockdown speed s with respect to changes in GDP_{day} , the weight on economic cost. The higher the GDP_{day} , the more costly the lockdown in economic terms. Next, we examine the sensitivity of pt to the optimal lockdown speed. It is important to analyze because the probability of transmission varies by environment. For households, it ranges from 0.10-0.40. For shared indoor spaces, it ranges from 0.01-0.05 (Bulfone et al. [7]). Varying this parameter provides a view on how optimal lockdown speed varies between seasons. In winter seasons, pt is higher since most people are interacting indoors, while in summer it is the converse.

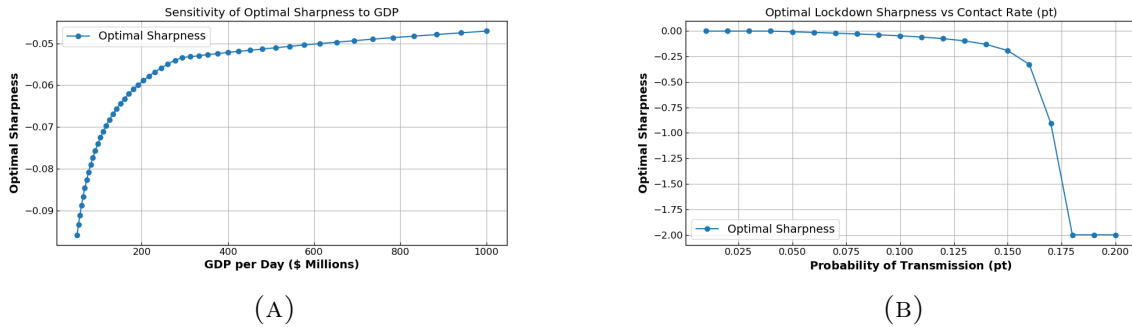


FIGURE 6. Sensitivity of Optimal Lockdown Speed to pt and α

Figure 6(a) shows the sensitivity of GDP_{day} . The range of values chosen for the sensitivity analysis of GDP_{day} was based on the GDP per capita averages across different regions of the U.S.. The conversion between GDP per day and GDP per capita is given by $GDP \text{ per capita} = \frac{GDP \text{ per day} * 365}{N}$.

At the lowest range, we have a GDP per day of 50 million, which translates to about \$9k GDP per capita, close to many rural areas in the U.S.. At the upper range, we have \$182k GDP per capita, which is close to high productivity areas like Silicon Valley and parts of New York City (Wikipedia contributors [27]). In Figure 4(a), we see that as GDP_{day} increases, the optimal lockdown speed slows. This makes sense because as one values the economy more, lockdowns become less favorable. Note, the shape of the

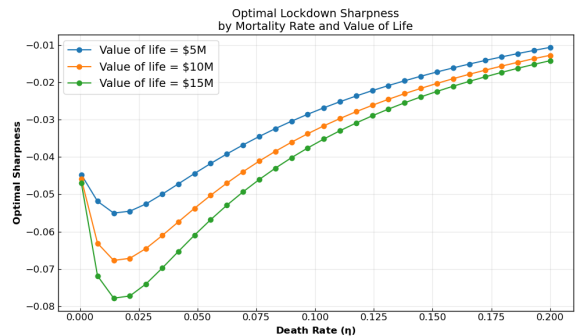


FIGURE 7. Sensitivity of Optimal Lockdown Speed to Death Rate (η)

curve is related to the curvature of the lockdown function and does not necessarily reflect diminishing returns to the parameter.

Turning our attention to Figure 6(B), we see that for low values of the parameter pt , the optimal speed is very slow. For a low probability of transmission, a quick lockdown isn't necessary. As pt increases, the lockdowns become quicker. The optimal s takes on the maximum boundary value of -2. This is intuitive because if a virus is spreading quickly, it's best to enact a swift lockdown.

Finally, we present a more interesting case, the sensitivity of optimal lockdown speed to death rate η . From Figure 6, the relationship between death rate and the optimal lockdown speed is not monotonic, as it was the case for the prior two parameters. The intuition is that for low death rates, a swift lockdown is preferred. However, as the death rate increases, a lockdown can only do so much to prevent deaths; therefore, the objective function starts to prefer the economy over lives when pt is large. The curves in Figure 6 also vary by the value of life. As the value of life increases, the objective function prefers sharper lockdowns, confirming intuition.

4. DISCUSSION

For our analyses, we fixed the initial condition to start with a single exposed individual, a patient zero scenario. We argue that starting with this initial condition is the most realistic because a location can't immediately spawn ten exposed individuals unless there is travel, but the model does not take this into account. The model is meant for a single environment. N is fixed at 2×10^6 . Code was modified to vary this value and we found that changing the value doesn't change the outcome. We acknowledge that the value of a human life is subjective, hence the sensitivity analysis, which allows it to range from \$5M to \$15M.

An exploration of adding additional parameters was conducted, such as the length of the lockdown and varying the lockdown start time. However, the length of the lockdown was found to be collinear with the speed of the lockdown. This makes sense given that a slower lockdown implies a longer, dragged-out lockdown period. Varying the start of the lockdown didn't yield interesting results because the optimizer always prefers to choose the earliest lockdown time.

Most results are intuitive and monotonic in relation; however, the sensitivity analysis reveals an interesting result for the death rate. Lockdowns are only effective up to a point. When death rates are very high, lockdowns are ineffective. In our simple objective function, it would be better to allow unavoidable deaths in favor of saving the economy.

Overall, the model achieved what we hoped to accomplish. It allowed us to model vaccinations, mutations, and the latent period. It also allowed us to determine different optimal lockdown policies, balancing death, economic cost, and hospital overcapacity. We even went as far as to model state-specific productivity to analyze how it changes during a lockdown.

5. CONCLUSION

Fortunately for us, the days of the pandemic are over, but it is still worthwhile to analyze it to learn from history. The model reveals a great deal about the dynamics between lockdown policies and the SEIRD model dynamics. We successfully extended the SIR to include two new states to better capture the nuances of a virus. We then presented a time-varying functional form for number of contacts modulated by a lockdown policy (s, m_{min}) . We presented an objective function that captures the total cost of a lockdown, then relaxed its assumptions to analyze state-specific productivity during a lockdown.

The sensitivity analysis revealed the limitations of a lockdown. If death rates are too high, there is only so much a lockdown can do. If a virus mutates, then a lockdown can not achieve full virus eradication. We acknowledge that our model doesn't capture the complete picture of the pandemic. Modifications to include factors such as age-dependent mortality rates or conducting additional sensitivity analysis on parameters are all viable extensions to the model.

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6. STATEMENT OF INDIVIDUAL CONTRIBUTION

Kaylee

- Worked on the SEIRD model, the optimization problem and sensitivity analysis.
- Wrote intro, description of the model, the first section of the analysis, discussion and conclusion.

Tadhg

- Worked on the economic productivity model
- Wrote the second section of the analysis
- Built the PowerPoint